16 According to government statistics, 6.45% of people in Britain are vegetarian.

A researcher believes the percentage of people who are vegetarian in the village where she lives might be different.

There are 484 people living in this village.

The researcher randomly selects 78 people from the village and establishes that 12 of them are vegetarian.

(a) Explain whether the 78 people selected form a sample or a population. **[1]**

(b) Conduct a hypothesis test at the 5% level to determine whether there is any evidence to support the researcher's belief that the percentage of people in her village who are vegetarian is different to 6.45%. **[7]**

(c) State the probability that the null hypothesis is incorrectly rejected. **[1]**